Tutorial No:5

Tutorial Title: Implement Stack and Queue using arrays.

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Program :

```
// Stack using array
#include <stdio.h>
#include <stdlib.h>
#define SIZE 5
struct stack
{
int s[SIZE];
int top;
} st;
int isFull()
{
if (st.top >= SIZE - 1)
return 1;
else
return 0;
}
int isEmpty()
{
if (st.top == -1)
return 1;
else
return 0;
}
int pop()
{
int item;
item = st.s[st.top];
st.top--;
return item;
}
void push(int item)
{
st.top++;
st.s[st.top] = item;
```

```
}
void display()
{
int i;

if (isEmpty())
printf("\nStack is empty!");
else
{
for (i = st.top; i >= 0; i--)
printf("\n%d", st.s[i]);
}
}
int main(void)
{
int item, choice;
char ans;
st.top = -1;
printf("\nImplementation of Stack");
do
{
printf("\nMain menu");
printf("\n1. Push\n2. Pop\n3. Display\n4. Exit");
printf("\nEnter your choice: ");
scanf("%d", &choice);
switch (choice)
{
case 1:
printf("\nEnter the item to be pushed: ");
scanf("%d", &item);
if (isFull())
printf("\nStack is full");
else
push(item);
break;
case 2:
if (isEmpty())
printf("\nEmpty stack (Underflow)");
else
{
item = pop();
printf("\nThe popped element is: %d", item);
}
break;
```

```
case 3:
display();
break;
case 4:
exit(0);
}
printf("\nDo you want to continue? (Y/N): ");
scanf(" %c", &ans);
} while (ans == 'Y' || ans == 'y');
return 0;
}
```

Output:-

Output

```
Implementation of Stack
Main menu
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 1

Enter the item to be pushed: 13

Do you want to continue? (Y/N): Y
```

Output

```
Main menu
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 2

The popped element is: 13
Do you want to continue? (Y/N): Y
```

```
Main menu
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 3

Stack is empty!
Do you want to continue? (Y/N): Y
```

```
Main menu
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 4
```

Code:-

```
//QUEUE using array
#include <stdio.h>
#include <stdlib.h>
#define SIZE 5
struct Queue {
int arr[SIZE];
int front, rear;
} q;
int isfull() {
if (q.rear >= SIZE - 1)
return 1;
else
return 0;
}
void insert(int item) {
if (isfull()) {
printf("\n Queue is full");
} else {
q.rear = q.rear + 1;
q.arr[q.rear] = item;
}
}
int isempty() {
if (q.front > q.rear)
return 1;
else
return 0;
}
int delete() {
int item;
if (isempty()) {
printf("\n Queue is Empty");
return -1;
} else {
item = q.arr[q.front];

q.front = q.front + 1;
return item;
}
}
void display() {
```

```
int i;
if (isempty()) {
printf("\n Queue is Empty");
} else {
for (i = q.front; i <= q.rear; i++) {
printf("%d\t", q.arr[i]);
}
}
}
void main() {
int choice, item;
q.front = 0;
q.rear = -1;
while (1) {
printf("\n Implementation of queue");
printf("\n 1. Insert");
printf("\n 2. Delete");
printf("\n 3. Display");
printf("\n 4. Exit");
printf("\n Enter Choice: ");
scanf("%d", &choice);
switch (choice) {
case 1:
printf("\n Enter item to insert: ");
scanf("%d", &item);
insert(item);
break;
case 2:
if (isempty()) {
printf("\n Queue is Empty");

} else {
item = delete();
}
break;
case 3:
display();
break;
case 4:
printf("\n Exiting the program...");
exit(0);
default:
printf("\n Invalid Choice !!");
}
```

}
}
Output:-

Output

```
Implementation of queue
1. Insert
2. Delete
3. Display
4. Exit
Enter Choice: 1

Enter item to insert: 13

Implementation of queue
1. Insert
2. Delete
3. Display
4. Exit
Enter Choice: 2
```

```
Implementation of queue
1. Insert
2. Delete
3. Display
4. Exit
Enter Choice: 3

Queue is Empty
Implementation of queue
1. Insert
2. Delete
3. Display
4. Exit
Enter Choice: 4

Exiting the program...
```